

Python Projects Report

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DECLARATION

I hereby declare that this Python Projects Report is the outcome of my own effort and work. It has not been submitted, either partially or entirely, to any other institution or for any other purpose. All references and sources used have been duly acknowledged.

Table of Contents

Acknowledgment

Declaration

Abstract

1. Introduction

2. Objective

3. Software & Tools Used

4. Program 1: Arithmetic and Quadratic Operations

5. Program 2: Linear Equation Solver

6. Program 3: Graph Plotting

7. Program 4: Python Function

8. Program 5: GUI Application using
Tkinter

9. Conclusion

Abstract

This report presents a series of Python-based programming tasks focused on mathematical computations and graphical interface design. The lab includes arithmetic and algebraic implementations, visual representation using graphs, recursive function logic, and simple GUI development with Tkinter. The goal is to enhance problem-solving skills and understand Python's application in various domains such as mathematics, data visualization, and user interface development.

Introduction

Python is a high-level, interpreted programming language known for its simplicity and versatility. This lab assignment is designed to apply Python programming in solving mathematical problems and developing simple applications. The tasks include arithmetic operations, solving equations, visualizing functions using graphs, recursive functions, and creating GUI applications using Tkinter.

Objective

- To implement basic arithmetic and algebraic expressions using Python.
- To solve and visualize linear and quadratic equations.
- To understand and implement mathematical functions (like factorial).
- To create simple GUI applications using the Tkinter module.
- To develop and present these solutions in a readable and organized report format.

Software Tools Used

- **Python 3.x** – For writing and executing code
- **VS Code / IDLE** – Code editor/IDE
- **Matplotlib & NumPy** – For plotting graphs and handling numerical data
- **Tkinter** – For creating GUI applications
- **GitHub** – For uploading and sharing the project

Q1: Using Python implementation of any arithmetic operation or quadratic expression.

Arithmetic Operations

```
def arithmetic_operations(a, b):  
    print("Arithmetic Operations:")  
    print(f"{a} + {b} = {a + b}")  
    print(f"{a} - {b} = {a - b}")  
    print(f"{a} * {b} = {a * b}")  
    if b != 0:  
        print(f"{a} / {b} = {a / b}")  
    else:  
        print("Division by zero is not allowed.")
```

Quadratic Expression:

```
def evaluate_quadratic(a, b, c, x):  
    result = a * x**2 + b * x + c  
    print("\nQuadratic Expression:")  
    print(f"{a}x2 + {b}x + {c} at x = {x} is {result}")
```

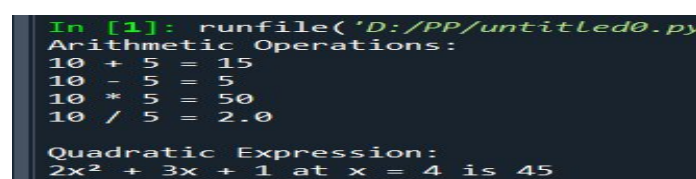
```
arithmetic_operations(10, 5)
```

```
evaluate_quadratic(2, 3, 1, 4)
```

. **arithmetic_operations(a, b)** performs and prints:

- Addition, subtraction, multiplication, and division of two numbers.
- Handles division by zero safely.

. **evaluate_quadratic(a, b, c, x)** calculates the value of a **quadratic expression**



```
In [1]: runfile('D:/PP/untitled0.py')  
Arithmetic Operations:  
10 + 5 = 15  
10 - 5 = 5  
10 * 5 = 50  
10 / 5 = 2.0  
  
Quadratic Expression:  
2x2 + 3x + 1 at x = 4 is 45
```

Q2: Implementation of Linear Equation

```
def linear_equation(m, x, c):
```

```
    y = m * x + c
```

```
    print(f"Linear Equation: y = {m}x + {c}")
```

```
    print(f"For x = {x}, y = {y}")
```

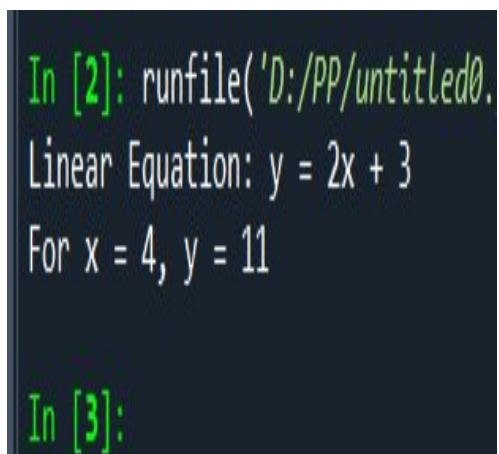
```
linear_equation(2, 4, 3)
```

. Defines a function to calculate the value of a linear equation:

$y=mx+c$

. Takes slope m, input x, and intercept c as inputs.

. Calculates y and prints the equation along with the result for the given x.



```
In [2]: runfile('D:/PP/untitled0.  
Linear Equation: y = 2x + 3  
For x = 4, y = 11  
  
In [3]:
```

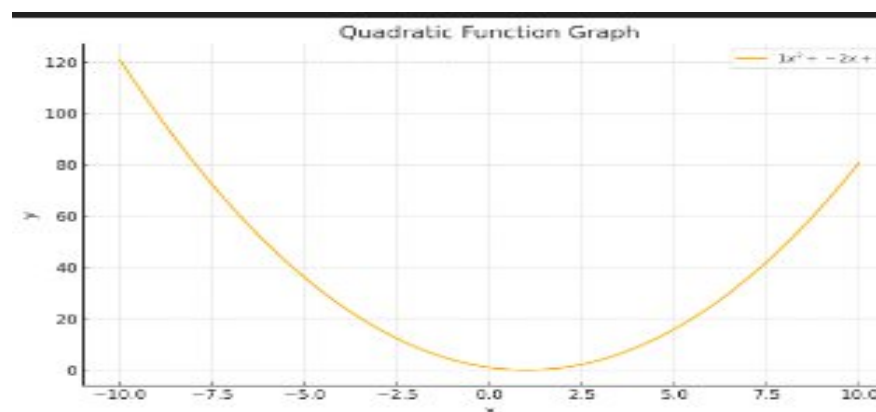

#Q3: Graphical Representation of a Mathematical Function

```
import matplotlib.pyplot as plt
import numpy as np

def plot_quadratic(a, b, c):
    x = np.linspace(-10, 10, 400)
    y = a * x**2 + b * x + c
    plt.plot(x, y, label=f'{a}x2 + {b}x + {c}')
    plt.title('Quadratic Function Graph')
    plt.xlabel('x')
    plt.ylabel('y')
    plt.grid(True)
    plt.legend()
    plt.show()
```

```
plot_quadratic(1, -2, 1)
```

- . Plots a quadratic equation: $y = ax^2 + bx + c$
- . Uses **NumPy** to generate x-values from -10 to 10
- . Calculates y-values using the quadratic formula
- . Uses **Matplotlib** to display the graph with title, labels, grid, and legend
- . `plot_quadratic(1, -2, 1)` plots: $y = x^2 - 2x + 1$



#Q4: Python Function Implementation

```
def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    return n * factorial(n - 1)  
  
num = 5  
  
print(f"Factorial of {num} is {factorial(num)}")
```

- . This function calculates the **factorial** of a number n using **recursion**.
- . **Factorial** means multiplying a number by all positive integers less than it.
- . $n! = n \times (n-1) \times (n-2) \times \dots \times 1$ $n! = n \times (n-1) \times (n-2) \times \dots \times 1$

```
In [1]: runfile('D:/PP/untitled0.py', wdir='D:/PP')  
Factorial of 5 is 120  
  
In [2]:
```

Q5: Using Tkinter to Make a GUI Application

```
import tkinter as tk

def add():
    result = int(entry1.get()) + int(entry2.get())
    label_result.config(text=f"Result: {result}")

root = tk.Tk()
root.title("Simple Adder")

entry1 = tk.Entry(root)
entry1.pack()

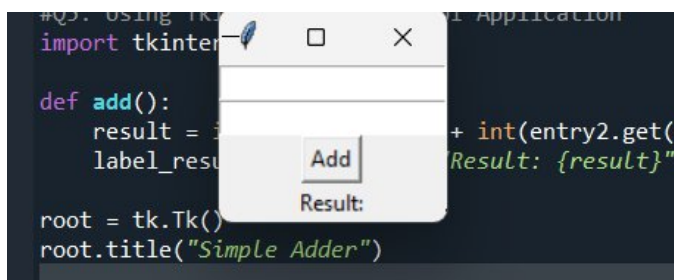
entry2 = tk.Entry(root)
entry2.pack()

button_add = tk.Button(root, text="Add", command=add)
button_add.pack()

label_result = tk.Label(root, text="Result:")
label_result.pack()

root.mainloop()
```

- . This Python code creates a **simple GUI app** using Tkinter.
- . It has **two input fields** for numbers and a **button** to add them.
- . When the **"Add" button** is clicked, it reads the inputs, adds them, and displays the **result**.
- . `root.mainloop()` keeps the window running.



Conclusion

This code showcases basic Python skills through arithmetic operations, quadratic and linear expressions, graph plotting with Matplotlib, a recursive factorial function, and a simple GUI adder using Tkinter. It combines math, visualization, and user interaction in a beginner-friendly way.

THANK YOU

